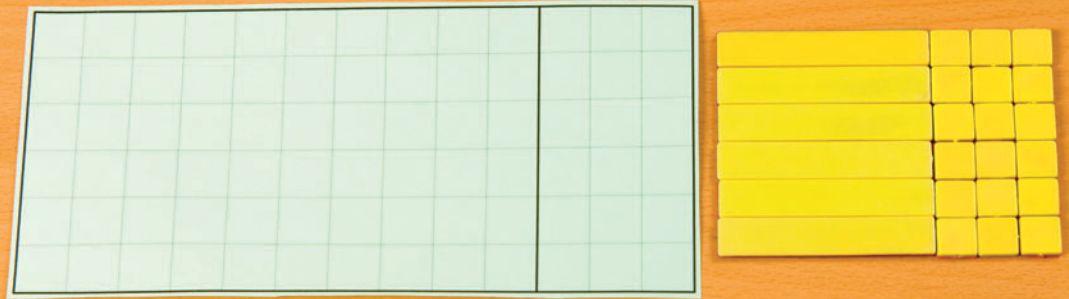


3.3 Common Factors of a Polynomial

LESSON FOCUS

Model and record factoring a polynomial.



Make Connections

Diagrams and models can be used to represent products.

What multiplication sentences are represented above?

What property do the diagrams illustrate?

Construct Understanding

The variable algebra tile can represent any variable we like. It is usually referred to as the x -tile.

THINK ABOUT IT

You may need algebra tiles.

Sketch all the ways you can arrange these tiles to form a rectangle.



Beside each sketch, write the multiplication sentence it represents.

Each set of tiles below represents the polynomial $4m + 12$.

The dimensions of each rectangle represent the *factors* of the polynomial.

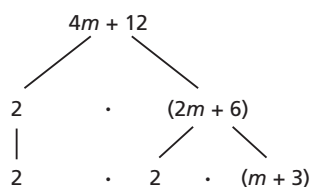
■ $1(4m + 12) = 4m + 12$



■ $2(2m + 6) = 4m + 12$



■ $4(m + 3) = 4m + 12$



When we write a polynomial as a product of factors, we *factor* the polynomial.

The diagrams above show that there are 3 ways to factor the expression $4m + 12$.

The first two ways: $4m + 12 = 1(4m + 12)$ and $4m + 12 = 2(2m + 6)$ are incomplete because the second factor in each case can be factored further.

That is, neither 1 nor 2 is the greatest common factor of $4m$ and 12.

The third way: $4m + 12 = 4(m + 3)$ is complete. The greatest common factor of $4m$ and 12 is 4.

We say that $4m + 12$ is *factored fully* when we write $4m + 12 = 4(m + 3)$; that is, the polynomial cannot be factored further.

Compare multiplying and factoring in arithmetic and algebra.

In Arithmetic	In Algebra
<i>Multiply</i> factors to form a product. $(4)(7) = 28$	<i>Expand</i> an expression to form a product. $3(2 - 5a) = 6 - 15a$
<i>Factor</i> a number by writing it as a product of factors. $28 = (4)(7)$	<i>Factor</i> a polynomial by writing it as a product of factors. $6 - 15a = 3(2 - 5a)$

Factoring and expanding are inverse processes.

Example 1**Using Algebra Tiles to Factor Binomials**

Factor each binomial.

a) $6n + 9$

b) $6c + 4c^2$

SOLUTIONS

a) $6n + 9$

Method 1

Arrange algebra tile in a rectangle.

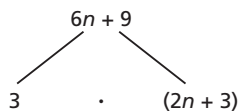
The dimensions of the rectangle are 3 and $2n + 3$.

So, $6n + 9 = 3(2n + 3)$

**Method 2**

Use a factor tree.

$6n + 9 = 3(2n + 3)$



b) $6c + 4c^2$

Method 1

Use algebra tiles.

$6c + 4c^2 = 2c(3 + 2c)$

**Method 2**

Use the greatest common factor.

Factor each term of the binomial.

$6c = 2 \cdot 3 \cdot c$

$4c^2 = 2 \cdot 2 \cdot c \cdot c$

The greatest common factor is $2c$.

Write each term as a product of $2c$ and another monomial.

$6c + 4c^2 = 2c(3) + 2c(2c)$

Use the distributive property to write the sum as a product.

$6c + 4c^2 = 2c(3 + 2c)$

CHECK YOUR UNDERSTANDING

1. Factor each binomial.

a) $3g + 6$ b) $8d + 12d^2$

[Answers: a) $3(g + 2)$

b) $4d(2 + 3d)$]

How can you check that your answer is correct?

When a polynomial has negative terms or 3 different terms, we cannot remove a common factor by arranging the tiles as a rectangle. Instead, we can sometimes arrange the tiles into equal groups.

Example 2 Factoring Trinomials

Factor the trinomial $5 - 10z - 5z^2$.

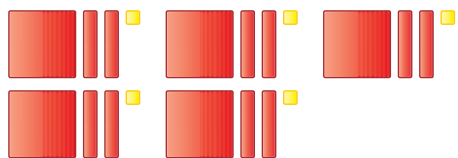
Verify that the factors are correct.

SOLUTIONS

$$5 - 10z - 5z^2$$

Method 1

Use algebra tiles. Arrange five 1-tiles, 10 negative z -tiles, and 5 negative z^2 -tiles into equal groups.



There are 5 equal groups and each group contains the trinomial $1 - 2z - z^2$.

So, the factors are 5 and $1 - 2z - z^2$.

$$5 - 10z - 5z^2 = 5(1 - 2z - z^2)$$

Method 2

Use the greatest common factor.
Factor each term of the trinomial.

$$5 = 5$$

$$10z = 2 \cdot 5 \cdot z$$

$$5z^2 = 5 \cdot z \cdot z$$

The greatest common factor is 5. Write each term as a product of the greatest common factor and another monomial.

$$5 - 10z - 5z^2 = 5(1) - 5(2z) - 5(z^2)$$

Use the distributive property to write the expression as a product.

$$5 - 10z - 5z^2 = 5(1 - 2z - z^2)$$

$$\begin{aligned} \text{Check: Expand } 5(1 - 2z - z^2) &= 5(1) - 5(2z) - 5(z^2) \\ &= 5 - 10z - 5z^2 \end{aligned}$$

This trinomial is the same as the original trinomial, so the factors are correct.

CHECK YOUR UNDERSTANDING

- Factor the trinomial $6 - 12z + 18z^2$.
Verify that the factors are correct.

[Answer: $6(1 - 2z + 3z^2)$]

When you have factored an expression, why should you always use the distributive property to expand the product?

Another strategy used in factoring polynomials is to identify and factor out any common factors.

Example 3 Factoring Polynomials in More than One Variable

Factor the trinomial. Verify that the factors are correct.

$$-12x^3y - 20xy^2 - 16x^2y^2$$

SOLUTION

$$-12x^3y - 20xy^2 - 16x^2y^2$$

Factor each term of the trinomial.

$$12x^3y = 2 \cdot 2 \cdot 3 \cdot x \cdot x \cdot x \cdot y$$

$$20xy^2 = 2 \cdot 2 \cdot 5 \cdot x \cdot y \cdot y$$

$$16x^2y^2 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot x \cdot x \cdot y \cdot y$$

The greatest common factor is: $2 \cdot 2 \cdot x \cdot y = 4xy$

Write each term as a product of the greatest common factor and another monomial.

$$-12x^3y - 20xy^2 - 16x^2y^2$$

$$= 4xy(-3x^2) - 4xy(5y) - 4xy(4xy) \quad \text{Write the expression as a product.}$$

$$= 4xy(-3x^2 - 5y - 4xy)$$

$$= 4xy(-1)(3x^2 + 5y + 4xy)$$

$$= -4xy(3x^2 + 5y + 4xy)$$

Check: Expand $-4xy(3x^2 + 5y + 4xy)$

$$= (-4xy)(3x^2) + (-4xy)(5y) + (-4xy)(4xy)$$

$$= -12x^3y - 20xy^2 - 16x^2y^2$$

This trinomial is the same as the original trinomial, so the factors are correct.

CHECK YOUR UNDERSTANDING

3. Factor the trinomial. Verify that the factors are correct.
 $-20c^4d - 30c^3d^2 - 25cd$

[Answer: $-5cd(4c^3 + 6c^2d + 5)$]

What other strategies could you use?

In *Example 3*, the common factor -1 was removed because every term was negative. When we factor a polynomial that has negative terms, we usually ensure that the first term inside the brackets is positive.

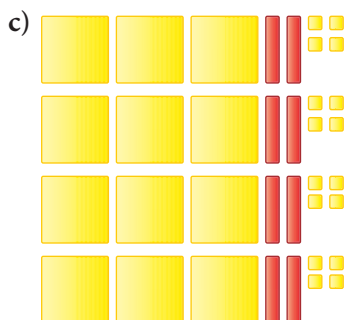
Discuss the Ideas

1. How is using an algebra tile model to multiply a polynomial by a monomial like using an area model to multiply two whole numbers?
2. How are the strategies used to factor a polynomial like those used to determine the prime factorization of a whole number?
3. Can every binomial be factored? Explain.

Exercises

A

4. For each arrangement of algebra tiles, write the polynomial they represent and identify its factors.



5. Factor the terms in each set, then identify the greatest common factor.

a) 6, 15n

b) 4m, m²

6. Use the greatest common factors from question 5 to factor each expression.

a) i) 6 + 15n

ii) 6 - 15n

iii) 15n - 6

iv) -15n + 6

b) i) 4m + m²

ii) m² + 4m

iii) 4m - m²

iv) m² - 4m

B

7. Use algebra tiles to factor each binomial. Sketch the tiles you used.

a) 5y + 10

b) 6 + 12x²

c) 9k + 6

d) 4s² + 14s

e) y + y²

f) 3h + 7h²

8. Factor each binomial. Why can you not use algebra tiles? Check by expanding.

a) 9b² - 12b³

b) 48s³ - 12

c) -a² - a³

d) 3x² + 6x⁴

e) 8y³ - 12y

f) -7d - 14d⁴

9. Use algebra tiles to factor each trinomial. Sketch the tiles you used.

a) 3x² + 12x - 6

b) 4 - 6y - 8y²

c) -7m - 7m² - 14

d) 10n - 6 - 12n²

e) 8 + 10x + 6x²

f) -9 + 12b + 6b²

10. Factor each trinomial. Why can you not use algebra tiles? Check by expanding.

a) 5 + 15m² - 10m³

b) 27n + 36 - 18n³

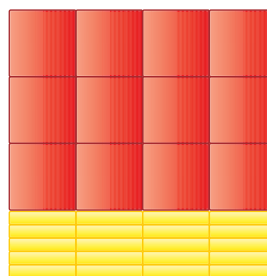
c) 6v⁴ + 7v - 8v³

d) -3c² - 13c⁴ - 12c³

e) 24x + 30x² - 12x⁴

f) s⁴ + s² - 4s

11. a) Write the polynomial these algebra tiles represent.



- b) Factor the polynomial.

- c) Compare the factors with the dimensions of the rectangle. What do you notice?

12. a) Here are a student's solutions for factoring polynomials. Identify the errors in each solution. Write a correct solution.

i) Factor: 3m² + 9m³ - 3m

Solution: 3m² + 9m³ - 3m = 3m(m + 3m²)

ii) Factor: -16 + 8n - 4n³

Solution: -16 + 8n - 4n³ = -4(4 + 2n + n²)

- b) What should the student have done to check his work?

13. Suppose you are writing each term of a polynomial as the product of a common factor and a monomial. When is the monomial 1? When is the monomial -1?

14. Simplify each expression by combining like terms, then factor.

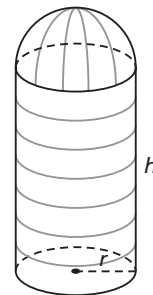
a) x² + 6x - 7 - x² - 2x + 3

b) 12m² - 24m - 3 + 4m² - 13

c) -7n³ - 5n² + 2n - n² - n³ - 12n

- 15. a)** Factor the terms in each set, then identify the greatest common factor.
- $4s^2t^2, 12s^2t^3, 36st^2$
 - $3a^3b, 8a^2b, 9a^4b$
 - $12x^3y^2, 12x^4y^3, 36x^2y^4$
- b)** Use the greatest common factors from part a to factor each trinomial.
- $4s^2t^2 + 12s^2t^3 + 36st^2$
 - $12s^2t^3 - 4s^2t^2 - 36st^2$
 - $-3a^3b - 9a^4b + 8a^2b$
 - $9a^4b + 3a^3b - 8a^2b$
 - $36x^2y^4 + 12x^3y^2 + 12x^4y^3$
 - $-36x^2y^4 - 12x^4y^3 - 12x^3y^2$
- 16.** Factor each trinomial. Check by expanding.
- $25xy + 15x^2 - 30x^2y^2$
 - $51m^2n + 39mn^2 - 72mn$
 - $9p^4q^2 - 6p^3q^3 + 12p^2q^4$
 - $10a^3b^2 + 12a^2b^4 - 5a^2b^2$
 - $12cd^2 - 8cd - 20c^2d$
 - $7r^3s^3 + 14r^2s^2 - 21rs^2$
- 17.** A formula for the surface area, SA , of a cylinder with base radius r and height h is:
- $$SA = 2\pi r^2 + 2\pi rh$$
- Factor this formula.
 - Use both forms of the formula to calculate the surface area of a cylinder with base radius 12 cm and height 23 cm. Is one form of the formula more efficient to use than the other? Explain.
- 18.** A formula for the surface area, SA , of a cone with slant height s and base radius r is:
- $$SA = \pi r^2 + \pi rs$$
- Factor this formula.
 - Use both forms of the formula to calculate the surface area of a cone with base radius 9 cm and slant height 15 cm. Is one form of the formula more efficient to use than the other? Explain.

- 19.** A silo has a cylindrical base with height h and radius r , and a hemispherical top.



- Write an expression for the surface area of the silo. Factor the expression.
Determine the surface area of the silo when its base radius is 6 m and the height of the cylinder is 10 m. Which form of the expression will you use? Explain why.
 - Write an expression for the volume of the silo. Factor the expression. Use the values of the radius and height from part a to calculate the volume of the silo. Which form of the expression will you use? Explain why.
- 20.** Suppose n is an integer. Is $n^2 - n$ always an integer? Justify your answer.
- C**
- 21.** A cylindrical bar has base radius r and height h . Only the curved surface of a cylindrical bar is to be painted.
- Write an expression for the fraction of the total surface area that will be painted.
 - Simplify the fraction.
- 22.** A diagonal of a polygon is a line segment joining non-adjacent vertices.
- How many diagonals can be drawn from one vertex of a pentagon? A hexagon?
 - Suppose the polygon has n sides. How many diagonals can be drawn from one vertex?
 - The total number of diagonals of a polygon with n sides is $\frac{n^2}{2} - \frac{3n}{2}$. Factor this formula. Explain why it is reasonable.

Reflect

If a polynomial factors as a product of a monomial and a polynomial, how can you tell when you have factored it fully?